Unit - I

Introduction to financial management

Financial management is consult with the effective use of and important economic resource. namely capital fund.

The financial management is an Area of financial decision making harmonizing individual motives and enterprise goals.

Effective utilization of resources following decisions has to be taken.

- Investment decision
- Financing decision
- Divident decision
- working capital management.

Functions of finance manager

- primary, secondary or subsidiary.

primary

- Estimating the requirements of fund (Financial forecasting, planning)
- pocurement of funds (raaustion of funds)
- Investment of funds.

Subsidiary

- helps in evaluation in decisions
- maintaining proper liquidity
- management of cash receivables and inventory.

- disposal of surplus.
- Performance evaluation and financial Control.

Objectives of financial Management.

Basic objective	Operational objective	Other objective.
1. profit maximization	1. Estimating the financial requirement	1. To serve the interests of Investors, owners, employees Etc.
2. Wealth maximization	2. acquiring sufficient & Timely funds.	2. Social welfare
	3. proper utilization of funds.	3. Economic welfare
	4. Safety of Investments	4. maintaining good relations with investors lenders Employees.
	5. Financial controls	5. To make timely payments of salaries, Interest, Tax Etc.
	6. Increasing profitability	
	7. Maximizing value	
	8. growth of the firm.	

Techniques of Time value of Money.

1. Compounding Technique
2. Discounting or present values techniques.

1) A person Being offered ₹100 Today may wait for a year if he ensured ₹ 110 at the End of one year (taking his prefrence of an interest of 10% p.A) The cashflow of ₹ 100 at present or ₹110 after one year will be the same for this person.

$V_1 = V_0 (1+i)$

V_1 = future value at the period I

V_0 = value of money at the time zero that is original sum of money.

I = Interest rate:

$V_1 = V_0 + (1+i)$

$V_1 = 100 + (1+1.1)$

$V_1 = 100 (1.1)$

$V_1 = 110$

$V_2 = 110 (1+0.1)$

$= 110 (1.1)$

$= 121$

$V_3 = 121 (1+0.1)$

$= 121 (1.1)$

$= 133.1$

(or)

$V_{10} = V_0 (1+i)^n$

$= ₹100 (1+0.1)^{10}$

$= ₹100 (1.1)^{10}$

$= ₹259.37$

$V_3 = V_0 (1+i)^n$

$= ₹100 (1+0.1)^3$

$= ₹133.1$

2. Umair Deposited ₹1 lakh in a savings Bank a/c at 5% S.I for a period of 5 years, what his accumulated Interest.

S.I = principle × Rate of Interest × per annum × no. of years.

S.I = P.T.R

$\Rightarrow 1,00,000 \times \frac{5}{100} \times 5$

$\Rightarrow 1,00,000 \times 0.05 \times 5$

$\Rightarrow 25000$

Compound factor table.

Period	% (10%)	
1	1.1	1 (1+0.1)
2	1.21	1.1 (1+0.1)
3	1.331	1.21 (1+0.1)
4	1.4641	1.331 (1+0.1)
5	1.61051	
6	1.7715	
7	1.948	
8	2.1435	

8.1

1 (1+0.08) = 1.08

1.08 (1+0.08) = 1.1664

1.1664 (1+0.08) = 1.259

1.259 (1+0.08) = 1.360

1.360 (1+0.08) = 1.469

1.469 (1+0.08) = 1.587

1.587 (1+0.8) = 1.71

1.71 (1+0.8) = 1.85

1.87 (1+0.8) = 1.999

4) If you deposit ₹5,000 today at 6% rate of Interest in how many years will this amount double.

Double period = $\frac{72}{6}$ = 12 years

5) multiple Compounding periods.

$V_n = V_0 \left(1+\frac{i}{m}\right)^{m \times n}$

V_n = future value of money after n years

V_0 = Value of money at time zero that is Original sum of money.

I = Interest Rate

m = no. of times of Compounding per year

n = no. of years.

Calculate the compound value of 10000 at the end of 3 years at 12% percent rate of Interest

(i) yearly basis

(ii) Quaterly

quaterly = $V_n = V_0 \left(1+\frac{i}{m}\right)^{m \times n}$

yearly = $V_0 = (1+i)^n$

yearly = $10000 \left(1+\frac{12}{1}\right)^{3 \times 1} = 14049$

quaterly = $10000 \left(1+\frac{0.12}{4}\right)^{4 \times 3}$

= 14257

Calculate the future value at the end of 5 years of the following series of payment at 10% Interest.

V Future value	I Years	II payments	III Compounded no. of years	IV Compounding Interest factors
1464	1	1000	4	1.4641 $(1+1.1)^4$
2662	2	2000	3	1.331 $(1+1.1)^3$
5633	3	3000	2	1.21 $(1+1.1)^2$
2200	4	2000	1	1.1 $(1+1.1)^1$
1500	5	1500	0	1 $(1+1.1)^0$
11456				

Payments × Compound Interest factor = future value

146C

Q) MR. A deposits ₹ 1000 at the end every year and the deposits earns a compound interest at the rate of 10% per annum. determine how much money now at the end of four years.

Year	payments	no of Yr Compounded	Compound Interest Yr	future value
1	1000	3	1.331	1331
2	1000	2	1.21	1210
3	1000	1	1.1	1100
4	1000	0	1	1000
			1000 ×	4641

At the end of 1 Yr ₹1 → @ 10% → 1 yr.

₹ 1.1

₹ 1.21

₹ 1.331

₹ 1.464

$V_n = V_0 (1+i)$

$\frac{V_n}{1+i} = V_0 = \frac{100}{1+0.1} = ₹\ 90.9.$

Present value or discounting 10%

Calculate the present value of the cashflow assuming the discount rate of 10%.

Year	Cashflow	present value factor @ 10%	present value of cashflow
1	₹ 5000	0.909	4545
2	₹ 10,000	0.8263	8263
3	₹ 10,000	0.7513	7513
4	₹ 3,000	0.6830	2049
5	₹ 2000	0.6209	1242
	Total present value of C.P		23,612

Calculate the present value of cashflows assuming the discount rate of 8%.

Year	Cashflow	present value factor @ 8%	present value of cashflow
1	10,000	$\frac{1}{1.08} = 0.9259$	9258
2	20,000	$\frac{0.925}{1.08}$ 0.8573	17146
3	10,000	0.7938	7938
4	5000	0.7350	3675
	Cashflow present value		38018

Year	Cashflow	present value factor @ 10%
1	5000	0.909
2	5000	0.8263
3	5000	0.7513
4	5000	0.6830
5	5000	0.6209
		3.795 × 5000

unit - 1

Simple Interest = principle × Rate of Interest × no. of years.